

Name- Bhumika Shrotriya

Course Name – Data Analytics With GenAI

Subject – SQL

Topic – SQL JOINS

MySQL Workbench

```
CREATE DATABASE SQL_JOINS_ASSIGNMENT;
```

```
USE SQL_JOINS_ASSIGNMENT;
```

-- CREATE TABLES

-- Table 1: Customers

```
CREATE TABLE Customers (  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(100),  
    City VARCHAR(100));
```

```
INSERT INTO Customers (CustomerID, CustomerName, City) VALUES
```

```
(1, 'John Smith', 'New York'),
```

```
(2, 'Mary Johnson', 'Chicago'),
```

```
(3, 'Peter Adams', 'Los Angeles'),
```

```
(4, 'Robert White', 'Houston'),
```

```
(5, 'Nancy Miller', 'Miami');
```

-- Table 2: Orders

```
CREATE TABLE Orders (  
    OrderID INT PRIMARY KEY,  
    CustomerID INT,  
    OrderDate DATE,  
    Amount DECIMAL(10, 2));
```

```
INSERT INTO Orders (OrderID, CustomerID, OrderDate, Amount) VALUES
```

```
(101, 1, '2024-10-01', 250),
```

```
(102, 2, '2024-10-05', 300),
```

```
(103, 1, '2024-10-07', 150),
```

```
(104, 6, '2024-10-10', 450),
```

```
(105, 3, '2024-10-12', 400);
```

-- Table 3: Payments

```
CREATE TABLE Payments (  
    PaymentID VARCHAR(10) PRIMARY KEY,  
    CustomerID INT,  
    PaymentDate DATE,  
    Amount DECIMAL(10, 2)  
);  
  
INSERT INTO Payments (PaymentID, CustomerID, PaymentDate, Amount) VALUES  
(  
    'P001', 1, '2024-10-02', 250),  
    ('P002', 2, '2024-10-06', 300),  
    ('P003', 3, '2024-10-11', 450),  
    ('P004', 4, '2024-10-15', 200);
```

-- Table 4: Employees

```
CREATE TABLE Employees (  
    EmployeeID INT PRIMARY KEY,  
    EmployeeName VARCHAR(100),  
    ManagerID INT  
);  
  
INSERT INTO Employees (EmployeeID, EmployeeName, ManagerID) VALUES  
(  
    1, 'Alex Green', NULL),  
    (2, 'Brian Lee', 1),  
    (3, 'Carol Ray', 1),  
    (4, 'Eva Smith', 2),  
    (5, 'David Kim', 2);
```

-- SOLUTIONS

-- Question 1: Retrieve all customers who have placed at least one order

-- Using INNER JOIN

```
SELECT DISTINCT c.CustomerID, c.CustomerName, c.City
FROM Customers c
INNER JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

-- Question 2: Retrieve all customers and their orders, including customers who have not placed any orders

-- Using LEFT JOIN

```
SELECT c.CustomerID, c.CustomerName, c.City, o.OrderID, o.OrderDate, o.Amount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

-- Question 3: Retrieve all orders and their corresponding customers, including orders placed by unknown customers

-- Using RIGHT JOIN or LEFT JOIN

```
SELECT c.CustomerID, c.CustomerName, c.City, o.OrderID, o.OrderDate, o.Amount
FROM Customers c
RIGHT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

-- Question 4: Display all customers and orders, whether matched or not

-- Using FULL OUTER JOIN (MySQL doesn't support FULL OUTER JOIN directly, so use UNION)

```
SELECT c.CustomerID, c.CustomerName, c.City, o.OrderID, o.OrderDate, o.Amount
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
UNION
SELECT c.CustomerID, c.CustomerName, c.City, o.OrderID, o.OrderDate, o.Amount
FROM Customers c
RIGHT JOIN Orders o
ON c.CustomerID = o.CustomerID;
```

-- Question 5: Find customers who have not placed any orders

-- Using LEFT JOIN with NULL check

```
SELECT c.CustomerID, c.CustomerName, c.City
FROM Customers c
LEFT JOIN Orders o
ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;
```

-- Question 6: Retrieve customers who made payments but did not place any orders

```
SELECT c.CustomerID, c.CustomerName, c.City
FROM Customers c
INNER JOIN Payments p
ON c.CustomerID = p.CustomerID
LEFT JOIN Orders o ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;
```

-- Question 7: Generate a list of all possible combinations between Customers and Orders

-- Using CROSS JOIN

```
SELECT c.CustomerID, c.CustomerName, c.City, o.OrderID, o.OrderDate, o.Amount  
FROM Customers c  
CROSS JOIN Orders o;
```

-- Question 8: Show all customers along with order and payment amounts in one table

```
SELECT  
    c.CustomerID,  
    c.CustomerName,  
    c.City,  
    o.OrderID,  
    o.OrderDate,  
    o.Amount AS OrderAmount,  
    p.PaymentID,  
    p.PaymentDate,  
    p.Amount AS PaymentAmount  
FROM Customers c  
LEFT JOIN Orders o  
ON c.CustomerID = o.CustomerID  
LEFT JOIN Payments p  
ON c.CustomerID = p.CustomerID;
```

-- Question 9: Retrieve all customers who have both placed orders and made payments

-- Using INNER JOIN on both tables

```
SELECT DISTINCT c.CustomerID, c.CustomerName, c.City
```

```
FROM Customers c
```

```
INNER JOIN Orders o
```

```
ON c.CustomerID = o.CustomerID
```

```
INNER JOIN Payments p
```

```
ON c.CustomerID = p.CustomerID;
```